

C

Adapted from materials from Dr. Carrier



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What languages do y’all know?

Why learn C?

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- Get some exposure to a “low-level” language

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- Bolster your understanding of what “higher-level” languages do

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- Get some exposure to a “low-level” language
- Bolster your understanding of what “higher-level” languages do
- ***Really understand memory management***

Language differences – Compiling and running code

Three main categories:

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1. Compiled

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Three main categories:

1. Compiled
2. Interpreted

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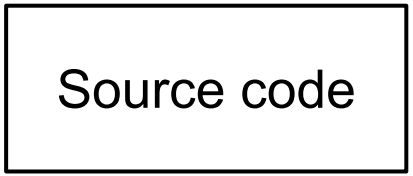
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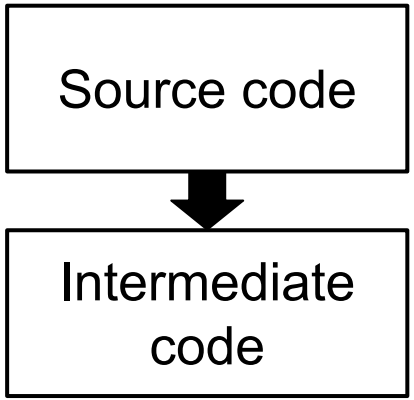
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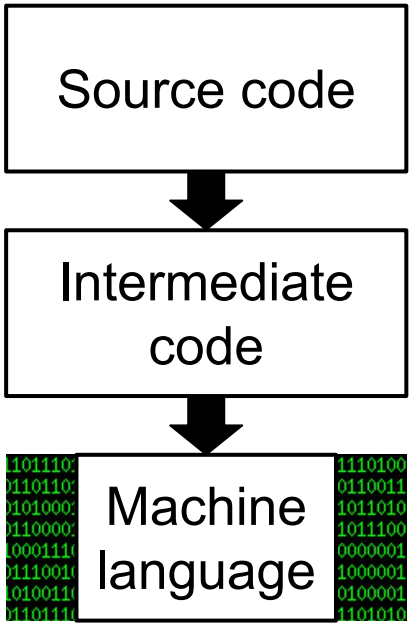
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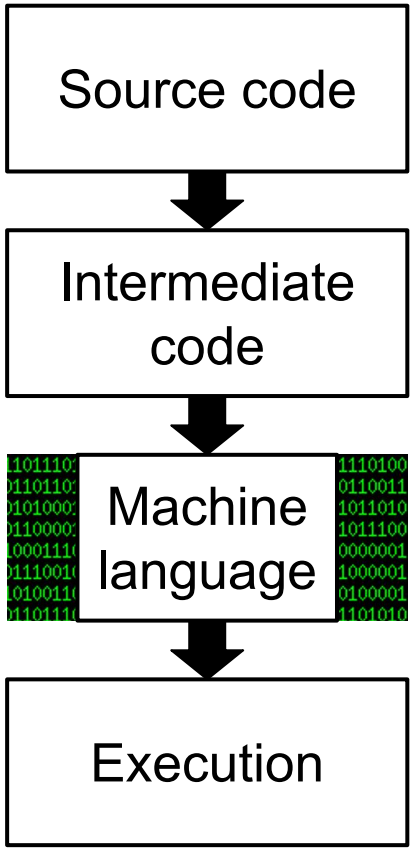
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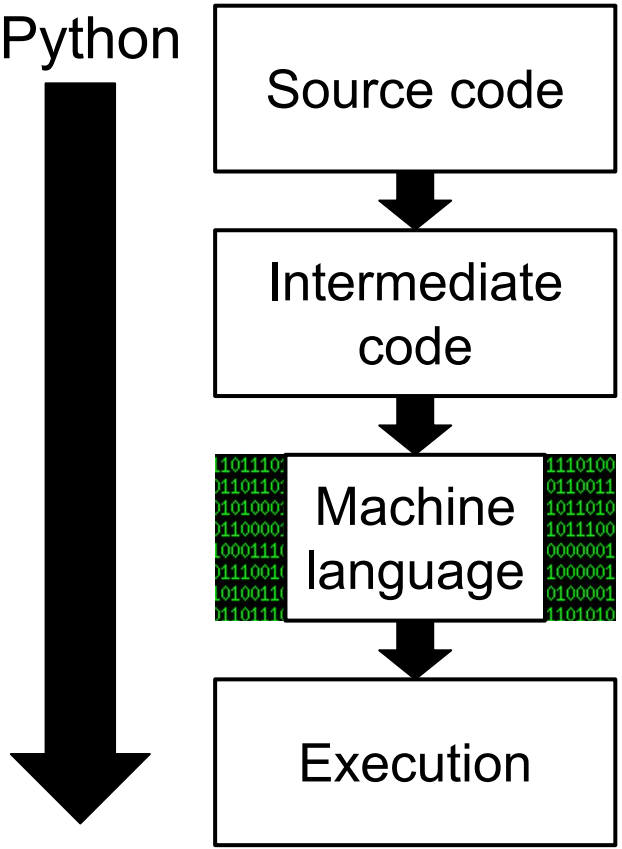
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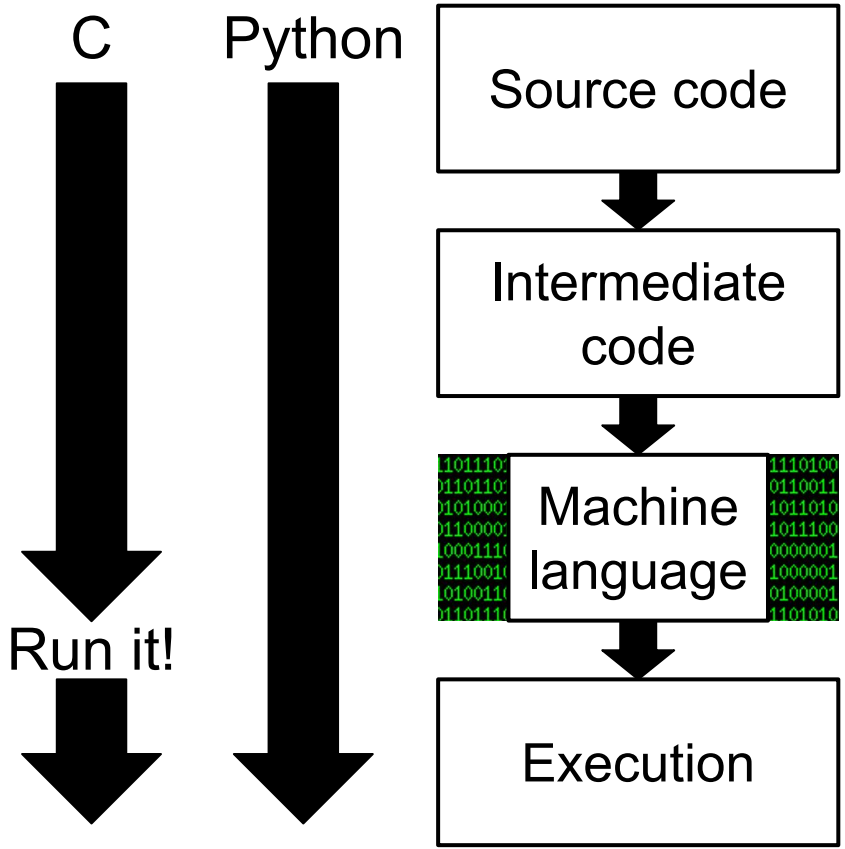
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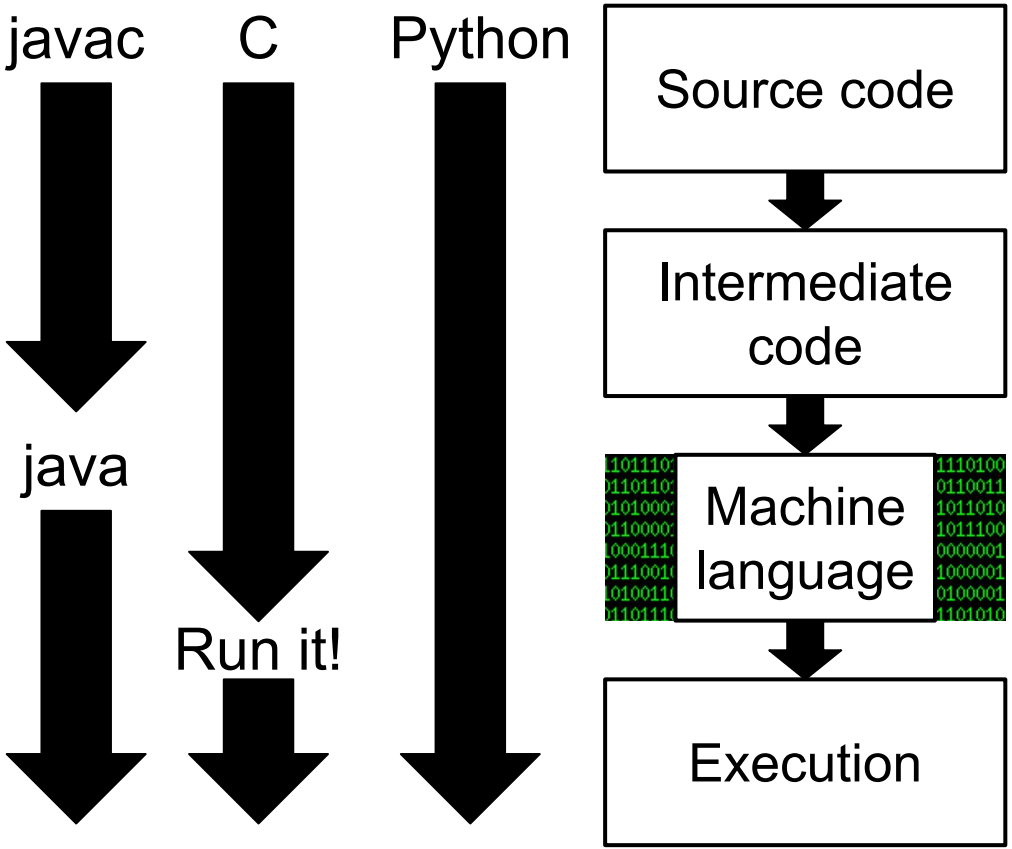
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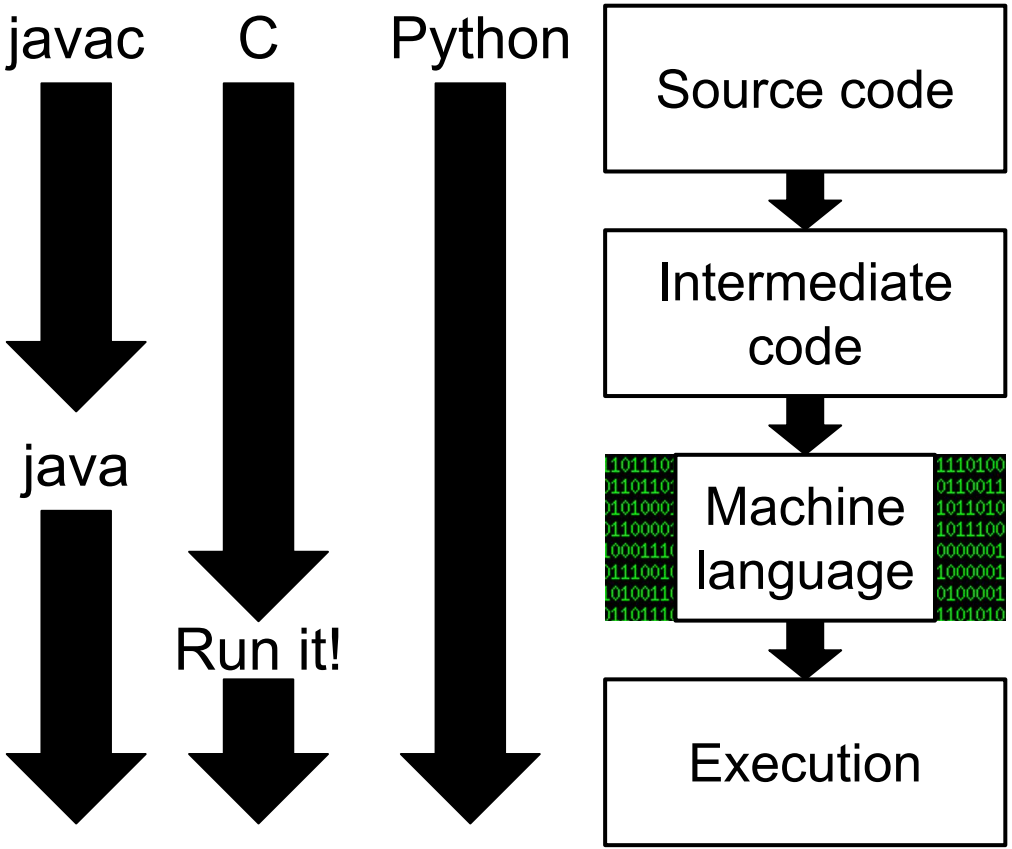


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What about other languages?



Language differences – Speed

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- The fast parts of Python are written in C/C++

Language differences – Speed

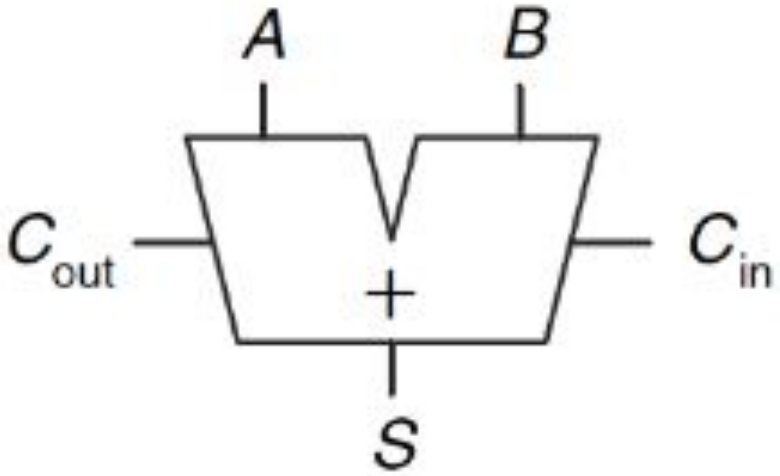
C is can be *way* faster than Python

- The fast parts of Python are written in C/C++
- Java falls between them, but can be very fast!

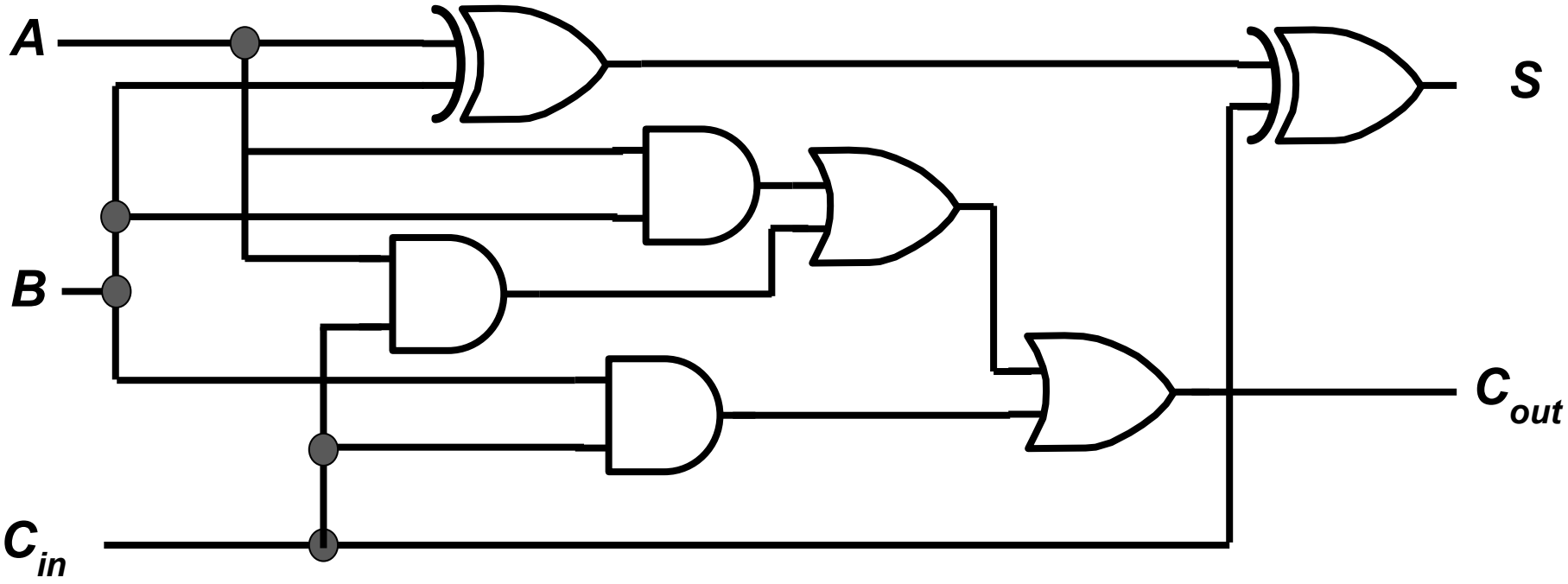
So why choose one over the other?

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- C++ is *almost* an extension to C
- C++ is designed to be object-oriented
  - C is “just” procedural
- Labels are confusing
- Understanding C will translate to C++
- Learning C++ kind of translates to C…

Why is it called C?

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Per wikipedia:

> A successor to the programming language B,